INT421—Digital Signal Processing

Final Exam Revision

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Final Exam Logistics

- Exam Time:
  1. January, 9th , 2025
  2. 3—Hour(s): from 10:00PM to 1:00PM
- Locations:
  1. Room [N247]: Students from 1:65
  2. Room [N311]: Students from 1:55
- Closed book, closed notes, clean desk.
- No lecture slides or textbooks allowed.
- I will not provide extra paper.

Exam Format

My Tests typically have the following format:

- True/False: 10 [**15 Marks**]
- Concepts and Definitions: 1 [**5 Marks**]
- Problems: 4 [**40 Marks**]
  1. Sampling
  2. Filter Realization
  3. Convolutions.
- Each question will indicate how many points it is worth (out of 60%)

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Study Topics

Will cover all material up to today...such as

- Signal properties
- System properties
- LTI Systems
- Difference equations
- Discrete-time convolution
- etc...

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Questions and Answers

Question

What is an advantage that digital signals have over analog signals?

Questions and Answers

Question

What is an advantage that digital signals have over analog signals?

1. They can eliminate noise that comes with the original signal and is allowed to perform operations on that signal
2. allow more noise to be heard
3. they can be stored easier
4. its easier to be amplified

Questions and Answers

Question

Determine the transfer function $H(z)$ for the following difference equation:

$$y[n] = y[n-1] + 2y[n-2] + 3y[n-3] + x[n-100]$$

Questions and Answers

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$$y[n] = y[n-1] + 2y[n-2] + 3y[n-3] + x[n-100]$$

$$Y(z)[1 - z^{-1} - 2z^{-2} - 3z^{-3}] = z^{-100}X(z)$$

$$\frac{Y(z)}{X(z)} = \frac{z^{-100}}{1 - z^{-1} - 2z^{-2} - 3z^{-3}}$$

Questions and Answers

Question

An FIR (finite impulse response) filter has an impulse response given by $h[n] = \{1, 2, 2, 3\}$. Find its response $y[n]$ to the input $x[n] = \{2, -1, 3\}$. Assume that both $x[n]$ and $h[n]$ start at $n = 0$.

Questions and Answers

Question

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The paper-and-pencil method expresses the input as: $x[n] = 2\delta[n] - \delta[n-1] + 3\delta[n-2]$ and tabulates the response to each impulse and the total response as follows

$h[n]$		$=$	1	2	2	3		
$x[n]$		$=$	2	-1	3			
Input	**Response**							
$2\delta[n]$	$2h[n]$	$=$	2	4	4	6		
$-\delta[n-1]$	$-h[n-1]$	$=$		-1	-2	-2	-3	
$3\delta[n-2]$	$3h[n-2]$	$=$			3	6	6	9
Sum $= x[n]$	Sum $= y[n]$	$=$	2	3	5	10	3	9

So, $y[n] = 2\delta[n] + 3\delta[n-1] + 5\delta[n-2] + 10\delta[n-3] + 3\delta[n=4] + 9\delta[n-5]$

Questions and Answers

Question

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Questions and Answers

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What is the Nyquist Theorem and why is it critical in digital signal processing?

The Nyquist Theorem, also known as the Nyquist-Shannon sampling theorem, states that a continuous signal can be completely reconstructed from its samples if it is sampled at a rate that is at least twice the maximum frequency component in the signal. This minimum sampling rate is called the Nyquist rate.

Why it is critical in DSP:

- Preventing Aliasing: It provides a criterion for selecting a sampling rate that can prevent aliasing, which is distortion caused by undersampling.
- Signal Reconstruction: It ensures that the original signal can be accurately reconstructed from its sampled version, preserving the integrity of the information contained in the signal.

Questions and Answers

Question

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$$H(z) = 1 - z^{-4}$$

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By inspection, the inverse Z-transform is $h[n] = \{1, 0, 0, 0, 1\}$

$$h[n] = \delta[n] + \delta[n-4]$$

Questions and Answers

Question

Define the discrete-time signal $x[n]$ as

$$x[n] = -0.3\delta[n+2] + 2.0\delta[n] + 1.5\delta[n-3] - \delta[n-5]$$

1. Sketch $x[n]$.
2. Write the Z-transform $X(z)$.
3. Define $G(z) = z^{-2}X[z]$. Sketch $g[n]$.

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Questions and Answers

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Think on it.

Questions and Answers

Question

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We know that

$$Z(nx[n]) \Longrightarrow -z\frac{dX(z)}{dz}$$

$$Y(z) = 5z^{-1} + 8z^{-2}$$

Questions and Answers

Question

What is the inverse Z-transform of the following function?

$$H(z) = \frac{z^{-2} - z^{-4}}{1 - z^{-1}}$$

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We can apply the Binomial theorem

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$$

to write

$$\begin{aligned} 1 - z^{-1} &= 1 + z^{-1} + z^{-2} + \dots \\ \Rightarrow H(z) &= \{z^{-2} - z^{-4}\}\{1 + z^{-1} + z^{-2} + \dots\} \\ &= z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6} \end{aligned}$$

Question

Find the transfer function $H(z)$ of the LSI system described by the following difference equation:

$$y[n] = -\frac{1}{2}y[n-2] + x[n] + x[n-1]$$

Questions and Answers

Question

Find the transfer function $H(z)$ of the LSI system described by the following difference equation:

$$y[n] = -\frac{1}{2}y[n-2] + x[n] + x[n-1]$$

We take the Z-transform of both sides and re-arrange as follows:

$$Y(z) = \frac{1}{2}z^{-2}Y(z) + X(z) + z^{-1}X(z)$$

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{1+z^{-1}}{1+\frac{1}{2}z^{-2}} = \frac{z^2+z}{z^2+\frac{1}{2}}$$

Questions and Answers

Question

Determine $h(n)$ for

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We have

$$\frac{H(z)}{z} = \frac{(z-4)}{(z-3)(z-2)} = \frac{A}{(z-3)} + \frac{B}{(z-2)}$$

Avery simple method to determine A and B is to multiply the equation by the denominator term $(z-3)(z-2)$. Thus we obtain

$$(z-4) = (z-3)B + (z-2)A$$

by setting $z = p_1 = 3$, we eliminate the term involving B. Then $A = -1$, similarly $B = 2$. This leads to

$$H(z) = -\frac{z}{(z-3)} + 2\frac{z}{(z-2)}$$

From this form we can quickly conclude that $h(n) = -(3)^n u[n] + 2(2)^n u[n$

Questions and Answers

Question

Given the first-order linear constant coefficient difference equation

$$y[n] = ay[n-1] + x[n]$$

find the impulse response of the system by first finding the system function $H(z)$ and then taking the Z-inverse transform. Assume a causal system.

Questions and Answers

Question

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find the impulse response of the system by first finding the system function $H(z)$ and then taking the Z-inverse transform. Assume a causal system.

Take the Z-transform of both sides ignoring initial conditions, collect terms, and divide $Y(z)$ by $X(z)$ to get $H(z)$ as follows:

$$Y(z) = az^{-1}Y(z) + X(z)$$
$$(1 - az^{-1})Y(z) = X(z)$$
$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - az^{-1}} = \frac{z}{1 - z}$$

By the causality requirement, the inverse transform of $z/(1-z)$ is a positive time exponential and with the region of convergence $|z| > |a|$, $h(n)$ becomes

$$h(n) = Z^{-1}\left(\frac{z}{1-z}\right) = a^n u(n)$$

Any Questions?

